Promo Finder: ranked promo cut lists from screenplays

Kuku AI Generalist Assignment · Problem 1, Find the Promos

Live app (Streamlit)	Code + run steps	Raw outputs	Longer explainer
https://promo-finder-demo.streamlit.app	github.com/Bhaypreet/promo_finder	Drive folder	Project write-up

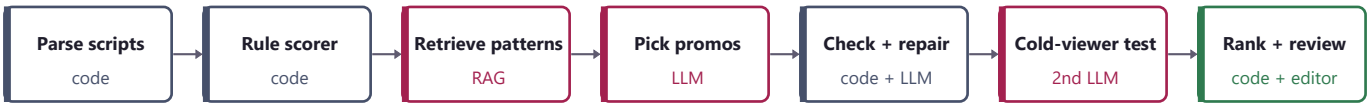
Give the tool a folder or .zip of episode screenplays. It returns a ranked list of 20-60 s promos. Each one comes with the episode and start/end time of every segment, the hook, the cliffhanger, a one-line reason it works, and its quality-check results, as a CSV an editor can cut from directly. On the full input pack it found 71 promos across 49 of the 50 episodes.

Problem and scope

Why Problem 1. It is a weekly, repeated cost: an editor watches every episode of every new show to find promo moments, and a new show ships every week. The screenplays already time-stamp every line, action and shot, so the hard part is the editorial judgement (which moment stops a scroll, where it should end), not the cutting. That splits cleanly: a language model for judgement, plain code for time and rules. It is also the one problem with Kuku's own content, so every output can be checked against the real show.

Where the brief was silent, I assumed	What the tool does about it
The script timestamps match the final cut.	Every cut snaps to the nearest shot change and never lands mid-line, so small drift stays invisible; an editor can nudge.
A promo's job is to bring **new** viewers to Episode 1.	Ranking gives early-episode promos a small bonus and penalises spoilers of late-series twists.
Platform is TikTok / Reels / Shorts, US audience.	The 3-second hook is checked from script timing. Visual impact (framing, faces) cannot be judged from text, so an editor watches the shortlist.
A promo can join several moments, from one or many episodes.	Segments must stay in story order, must not overlap and must total 20-60 s.
Free tiers only.	Token-lean prompts on Groq's free models. Any stronger model, including Kuku's own APIs, is a one-flag swap.

How it works



	Given to the language model	Kept as code or simple rules	Kept with a human
Work	Choosing the moments; writing the hook, cliffhanger and why; fixing a pick the checker rejected; judging a clip cold, as a stranger would	Reading the scripts; every timestamp; shot-cut snapping; the 20-60 s and 3 s rules; candidate windows; retrieval; the ranking formula	Final say on the top 10 (review mode); approved promos are saved and guide future shows
Why	Judgement over story and emotion is what models do well	Arithmetic and hard rules must be exact and repeatable	Taste, brand fit and spoiler calls

Key design decisions

Decision	Why it matters
The model picks line IDs, never timestamps. Each line gets an ID (B031); code turns IDs into times.	Models invent times but reference IDs reliably. Cuts land on shot changes and never mid-sentence.
Validated output with a repair loop. Answers must fit a strict schema; a pick that breaks a rule goes back to the model once, with the exact error.	14 of 71 first picks broke a rule (wrong length, overlapping segments, a late hook) and came back fixed; code fixed 3 more on its own.
A playbook the model retrieves from (RAG). 16 proven promo patterns (public humiliation, interrupted wedding, hidden identity) plus every promo an editor approves.	Same model, with vs without the playbook: precision against reference promos rose from 0.55 to 0.88. The tool gets better with each approved promo.
A second, different model judges every clip cold. It sees only the clip, with no show context.	The picker cannot grade its own work. This 'would a stranger stop and understand?' score is 40% of the rank.
Search, don't stuff, for cross-episode promos. 1,640 short moments are indexed for the whole show.	The full show is about 45K tokens. Retrieval finds matching moments across episodes and keeps prompts small.
One setting picks the model; every answer is cached.	Switching to a stronger or in-house model is one flag. Reruns cost nothing and give identical output.

Scale and cost

- **Cost per show** (50 episodes): 142 model calls, about 267K tokens. $0 on Groq's free tier; under $0.10 at Groq's published paid rates. The rules-only mode is free and runs in under a second.
- **Time:** 47 minutes on the free tier, set by its rate limits (about 8K tokens a minute), not by compute. Episodes are processed in parallel, so a paid tier or an in-house model cuts this to minutes.
- **Every show, every week:** a FastAPI service with a Dockerfile, plus the hosted demo. Upload a zip, get the cut list. Model answers and the search index are cached, so reruns are free.
- **Editor time** goes from watching every episode to reviewing a ranked top 10. **Next measure:** each posted promo's 3-second view rate, click-through and Episode 1-to-5 retention, used to re-weight the ranking.

Top promos and what I would post first

Post	Promo	Episode and cut	Sec	Opens on (first 3 s)	Ends on (cliffhanger)	Rank
1	**Disowned at the church door**	Ep 5 · 00:06.6-00:57.8	51	"You will not leave this church carrying the Ashcroft name."	Her mother's ledger hits the mud; she kneels in the rain.	#2
2	**Jilted at the altar**	Ep 1 · 00:17.8-01:12.4	55	"Not anymore." The groom is marrying her sister.	"And you knew?" (to her sister)	#5
3	**Night fire rescue**	Ep 18 · 00:19.3-00:48.8 + 02:37.8-03:06.5	58	Bedroom door kicked in, hallway in flames.	"What are you going to do?" Her husband's cold stare.	#3
4	**The branding iron**	Ep 32 · 01:59.5-02:34.4	35	"Silence! Silence!" in a packed courtroom.	The glowing iron hovers inches from her eye.	#4
5	**Crawling through thorns**	Ep 13 · 00:47.0-00:56.8 + 01:11.2-01:47.0	46	"Juliana, stop!"	Her bloody hand closes on the lost ring.	#6
6	**"Wife?!" The Duke reveal**	Ep 45 · 01:08.6-01:20.5 + 01:26.3-01:53.2	39	"Wife?!" He names her his Duchess.	Priest: "You are not his lawful wife."	#1

- **Post first: the premise.** Episodes 5 and 1 set up the whole show (jilted at the altar, then cast out). A cold viewer understood both (8/10), and they send new viewers to the start, where the payoff is immediate.
- **Hold the #1 pick for retargeting.** Episode 45 scores highest on every signal, but it gives away the title twist. It works best on people already watching. Catching this is the editor's job in review mode.

Quality checks

Check	What it tests	Result on the full pack
Hard rules (code, every promo)	20-60 s, hook in the first 3 s, ends on the cliffhanger, story order, no mid-line cuts	**69 of 71 pass** (18-59 s, mean 37 s). The 2 that miss (under 20 s by 0.4 s and 2.3 s) are flagged and ranked last.
Cold viewer (different model, clip text only)	Would you stop scrolling? Want the next episode? Understand it without context?	Top 10: **7.1 / 7.8 / 6.1** out of 10. Whole list: 4.0 / 4.9 / 4.1; clips that need context sink in the ranking.
Reference promos	13 promos marked by hand in 8 episodes before seeing the tool's output; a match is at least 40% time overlap	See the table below.

Run, against the 13 reference promos	Found (recall)	On target (precision)	Opens on the same hook
Random 30-45 s windows, 2 per episode (floor)	0.60	0.61	0.22
Rule scorer alone (free, no model)	0.92	0.75	0.25
gpt-oss-20b, no playbook	0.46	0.55	0.17
gpt-oss-20b + playbook RAG	**0.54**	**0.88**	**0.43**

- **The playbook works.** On the same model it lifts all three numbers, and more than doubles how often the promo opens on the reference hook.
- **The rule scorer finds where the action is** (recall 0.92), which is why its candidate windows go to the model as hints. It cannot write the hook or why, judge whether a clip stands alone, join moments or flag spoilers; the model does that.
- **The main run (gpt-oss-120b)** matches fewer exact windows (recall 0.46, precision 0.64). It prefers longer, joined promos that open on a different hook a few seconds away; its Episode 45 pick opens on "Wife?!" 6 s before the reference cut, and the cold viewer rated it 8/9/9.

Limitations: the five that matter

Limitation	In simple words, with a real example	What I do about it
1. It reads text, not video	It cannot see acting, faces or framing. A slap that reads well on paper can look weak on screen.	An editor watches the top 10 before posting: about 10 minutes per show.
2. Many clips need context	Scenes start mid-story, so a stranger may not know who is who. Cold viewers understood clips 4.1/10 on average.	Understanding is 40% of the rank, so confusing clips sink (top 10 average 6.1/10).
3. It misses spoilers	It does not know which reveal is the show's big twist. The #1 promo (Ep 45) gives away the Duke twist, and the model rated the risk 'low'.	The editor decides. Next: a per-show 'never reveal' list the model must respect.
4. Cross-episode montages are weak	Joining moments from different episodes loses the thread. One demo montage scored 2/10 with the cold viewer.	Kept, but ranked low. Next: only join 2-3 nearby episodes.
5. My test set is small	13 reference promos picked by me in 8 episodes: one person's taste. The main model matched 46% of them, often choosing a different, also good, moment.	Next: real results from posted promos (3-second view rate, Episode 1 clicks) to tune the ranking.

Deployment, and what went wrong when I used it

I deployed the prototype on **Streamlit Community Cloud** (https://promo-finder-demo.streamlit.app), running the same pipeline as the command line. Using it myself showed two problems:

- **Checking the results by hand took a lot of time.** On the free tier an AI run takes about 4 minutes for the 3-episode demo and 47 minutes for the full pack. Then every promo had to be checked against the script: find the start and end time, confirm the hook and the cliffhanger line. Doing that for 71 promos is slow, so the checks code can do (length, 3 s hook, shot-change cuts, ends on the cliffhanger) run on every promo, and a person only reviews the ranked top 10. Next: show each promo's script lines in the app.
- **The app could get stuck on "Another run is in progress".** Only one AI run is allowed at a time, because every visitor shares one free-tier token budget. If someone refreshed the page mid-run, the old run held the lock and new visitors waited with no result. Fixed: rules-only mode never waits, a busy AI run says so instead of hanging, a stuck run is released after 12 minutes, and every model call times out after 90 s.

Tools used, and where they got it wrong

Building: Claude Code as a coding assistant for the parser, pipeline, evaluation scripts and first drafts of this doc. **Inside the tool:** Groq gpt-oss-120b picks promos (gpt-oss-20b as fallback), Groq Qwen3.8-27b is the cold viewer, FastEmbed bge-small powers retrieval. **Where they got it wrong:** the picker broke a hard rule in 14 of 71 first picks: clips from 13 s to 88 s, overlapping segments, a 'hook' 67 s into the clip; the validator caught every one. The cold viewer first scored almost every clip 1-2/10 until I gave it a scoring guide with examples.